Experiment 13 — Case Study on Low-Code Limitations

Aim

To study the limitations of low-code development through a case study of a Hotel Room Booking & Front Office Management system.

Theory

Low-code development allows applications to be created using visual development tools, predefined components and limited amounts of code. Although it speeds up development, it has certain limitations.

Case Study: Hotel Room Booking & Front Office Management

A hotel wants to manage guest details, room bookings, check-in and check-out schedules, meal plans, booking status and guest feedback using Zoho Creator.

Advantages of Low-Code Development

- Faster application development.
- Reduced amount of manual coding.
- Easier creation of booking forms and reports.
- Built-in workflow automation for booking confirmations and reminders.
- Easier maintenance for simple hotel management applications.

Limitations

1. **Platform dependency:** The hotel booking application depends on Zoho Creator's capabilities and policies.
2. **Customization limits:** Highly specialized booking features or complex room allocation algorithms may require external code or services.
3. **Integration challenges:** Connecting payment gateways, external reservation platforms, or third-party hotel systems may require APIs and authentication.
4. **Performance constraints:** Large numbers of bookings and complex reports may require careful optimization.
5. **Vendor lock-in:** Moving the application to another platform may require redesigning forms, reports, workflows, and data structures.
6. **Licensing limitations:** Some advanced features, user-management options, integrations, or automation capabilities may require a paid plan.
7. **Security responsibility:** Guest information, contact details, and identity documents require proper access control and data protection.

Possible Solutions

- Use efficient workflows and Deluge scripts.
- Integrate external services when necessary.

- Review platform limits and licensing requirements.
- Apply security, testing, and data validation practices.
- Use pro-code extensions for specialized requirements.

Result

The advantages and limitations of low-code development were studied using the Hotel Room Booking & Front Office Management system as a case study.